SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I VISHAL.V (192419033) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **VISHAL. V**

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

SLOUTIONS: -

1. Hospital Doctor Shift Assignment Using Backtracking

Introduction

A hospital needs to assign three doctors (D1, D2 and D3) to three shifts: Morning, Afternoon and Night. Each doctor must receive exactly one shift, and all the given conditions must be satisfied. Backtracking helps find a valid assignment by checking the constraints at each step.

Given Constraints

1. D1 cannot work the Night shift.
2. D2 must work before D3.
3. Only one doctor can work each shift.
4. D3 cannot work the Morning shift.
5. Every doctor must receive exactly one shift.

Steps Using Backtracking

Step 1: Assign D1 to the Morning shift.
This is valid because D1 is not assigned to Night.
Step 2: Assign D2 to the Afternoon shift.
D2 is assigned before D3, so this satisfies the second condition.
Step 3: Assign D3 to the Night shift.
D3 is not assigned to Morning, and each shift has only one doctor.

Final Valid Schedule

Shift	Assigned doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion

Backtracking finds a valid schedule by assigning shifts one by one and checking the constraints. If any assignment violates a condition, it goes back and tries another option. The final schedule satisfies all the given conditions.

2. Robot Navigation in a 5 × 5 Grid Using Search

Introduction

A robot must move from the Start (S) to the Goal (G) in a 5 × 5 grid. Some cells contain obstacles, so the robot cannot pass through them. Each movement has a cost of 1.
Breadth-First Search (BFS) can find the shortest path when all moves have equal cost.

Given Constraints

- The robot can move Up, Down, Left or Right.
- Each movement costs 1.
- Obstacles cannot be crossed.
- Manhattan distance is used to estimate the distance to the goal.

Manhattan Distance Formula

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Here, (x_1, y_1) is the robot's current position and (x_2, y_2) is the goal position.

Steps Using BFS

1. Place the starting position S in a queue.
2. Remove the first position from the queue.
3. Check its neighbouring cells in the order Up, Down, Left and Right.
4. Ignore obstacles and already visited cells.
5. Add valid neighbouring cells to the queue.
6. Continue until the goal G is reached.
7. Trace the previous positions to obtain the shortest path.

Example of Node Expansion

The following is an illustrative example because the question does not show the actual grid layout.

Step	Node explored	Action
1	S	Add valid neighbours
2	First neighbour	Check its neighbours
3	Next neighbour	Skip obstacles
4	Further nodes	Continue BFS
5	G	Stop and trace the path

Optimal Path and Cost

The optimal path is the shortest valid path from S to G without crossing obstacles.

$$\text{Path cost} = \text{Number of moves} \times 1$$

\The exact path and numerical cost cannot be calculated without the positions of S, G and the obstacles.
Note: BFS uses a FIFO queue and does not use Manhattan distance to prioritize nodes. If Manhattan distance must guide the search, A* search is more appropriate because it uses f(n)=g(n)+h(n).

Conclusion

BFS is suitable for finding the shortest path when every move has the same cost. Manhattan distance can guide the search more efficiently when using A*.

3. Autonomous Rescue Robot in a Disaster-Affected Building

Introduction

An autonomous rescue robot is used to find survivors inside a damaged building. It must avoid blocked paths, identify risky areas and reach the survivor at the lowest possible cost.

Since the robot cannot observe the entire building at once, it must explore the surroundings and update its knowledge as it moves.

A. Analysis of Constraints

1. Movement: The robot can move Up, Down, Left or Right.
2. Obstacles: Blocked cells cannot be entered.
3. Limited sensing: The robot can observe only nearby cells.
4. Movement cost: Each normal move costs 1.
5. Risky zones: Entering a risky zone adds an extra cost of 2.
6. Minimum total cost: The robot must reach the survivor safely while reducing the overall cost.
7. Dynamic updates: The robot must update its map when it discovers new obstacles or risky areas.

These conditions mean the robot must consider both distance and safety rather than simply choosing the shortest route.

B. Solution Using Uniform Cost Search (UCS)

Uniform Cost Search is suitable because different paths may have different costs.

Steps:

1. Start at position S with a total cost of 0.
2. Add S to a priority queue.
3. Explore the available neighbouring cells.
4. Calculate the cost of each move:
    - Normal cell: \(1\)
    - Risky cell: \(1+2=3\)
    - Blocked cell: Cannot be entered.
5. Add valid positions to the priority queue according to their total path cost.
6. Update the robot's map whenever new information is discovered.
7. Continue until the survivor's position G is reached with the lowest possible path cost.

Cost calculation:

$$\text{Total cost} = \sum \text{movement costs}$$

For example, a route containing three normal moves and one risky move costs:

$$3(1) + 1(3) = 6$$

The actual optimal path depends on the locations of the obstacles, risky zones and survivor.

C. Application of AI Concepts

1. Intelligent Agent:

The rescue robot acts as an intelligent agent because it senses the environment, makes decisions and performs actions to reach the survivor.

2. Rationality:

A rational robot chooses the safest available route with the lowest total cost based on the information it has collected.

3. Environment Type

The environment is:

- Partially observable: The robot sees only nearby cells.
- Dynamic: New information may change its map.
- Sequential: Each move affects future decisions.
- Stochastic or uncertain: Hazards and unknown conditions may make outcomes uncertain.

D. Why UCS Is Suitable

UCS is suitable because it considers the total cost of a path rather than only the number of moves. It can avoid expensive risky routes when a cheaper safe route exists.

Since the robot initially has incomplete information, it should combine UCS with an online search approach. It explores, updates its map and replans when it discovers new obstacles or hazards.

Conclusion

An autonomous rescue robot can use UCS with online exploration to find a low-cost route to a survivor. By considering obstacles, risky zones and newly discovered information, it can make better decisions and navigate more effectively in a disaster-affected building.